DLSS 2025 - Assignment 4 Fine-tuning Large Language Models for Political Text Classification

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1 Introduction

The ParlaMint dataset contains speeches from various legislative bodies across multiple countries. For this report, we focus on parliamentary speeches delivered by members of the Austrian national council from 2017 to 2022. The objective of this assignment is to utilize and leverage a fine-tuned Large Language Model (LLM) to predict party affiliation and political orientation from political speeches. Between 2017 and 2022, Austria's political landscape was characterized by shifting coalitions, the rise of right-wing populism, and intense debates surrounding immigration, EU policy, and the COVID-19 response. These developments render Austrian parliamentary speeches from this period particularly valuable for analyzing ideological and partisan language patterns. To achieve this objective, we employ a Qwen 3.0 model with 0.6 billion parameters as our baseline and utilize LoRA (Low-Rank Adaptation) to fine-tune this model in order to obtain improved results

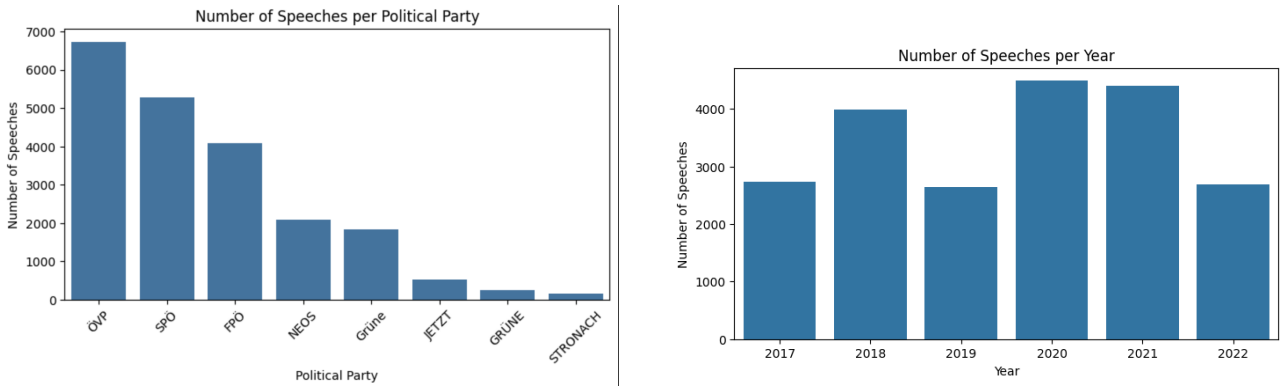
2 Results

2.1 Data

As mentioned above, we utilize the ParlaMint dataset, which is a collection of multilingual corpora of parliamentary speeches. In this report, we focus on Austria from 2017 to 2022. The dataset contains 256 parliamentary sessions, and after preprocessing steps, we worked with approximately 21,000 speeches. The data was divided into the actual speeches and metadata, and when processed correctly, each speech contains information on the speaker (name, birth, gender, role, status, and ID), the speaker's party (orientation and parliamentary status), and additional details (language, meeting, and legislative body). During preprocessing, we limited our analysis to speeches containing more than 50 words. Additionally, I filtered the data to exclude speeches about 'Regierungsgeschäfte' and removed speeches where the speaker's party or party orientation was missing. I also eliminated duplicate speeches. The results show that the ÖVP delivered the most speeches, followed by the SPÖ and FPÖ, ranging from 6,500 to 4,200 speeches within the selected timeframe. Other parties such as NEOS, GRÜNE (formerly JETZT), and STRONACH all contributed fewer than 2,000 speeches each. The distribution of speech length is right-skewed, varying between 50 and 2,000 words, with most speeches containing fewer than 1,000 words. The average speech length is 495.70 words, while the median length is 447.00 words. The years 2018, 2020, and 2021 recorded the highest number of speeches per year, whereas 2017, 2019, and 2022 showed lower counts. The increased number of speeches in 2020 and 2021 may be attributed to the COVID-19 pandemic, during which the National Council was required to convene more frequently to pass necessary legislation.

2.2 Methods

The baseline is a small Qwen 3 model with 0.6 billion parameters. It follows a standard causal decoder-only transformer architecture, where the model is trained to predict the next token given the previous ones. While this architecture is well-suited for tasks such as classification via prompting, out-of-the-box performance on specialized tasks like predicting political party affiliation or ideological orientation is limited. Initial classification results based on prompting alone were weak, likely due to the relatively small model size and the domain-specific nature of the task. It is plausible that larger models such as GPT-4 or Claude could perform better in zero-shot scenarios. However, this limitation provides strong motivation for domain adaptation via fine-tuning. The fine-tuning process employs Low-Rank Adaptation (LoRA). Using the LoRA approach, we do not modify all model weights but only small low-rank matrices, which results in faster, more efficient, and less computationally



(a) Speeches per party

(b) Speeches per year

Figure 1: Distribution of speeches across parties and years

expensive training by reducing the number of trainable parameters. For the first task, we reduced the number of trainable parameters to approximately 10 million parameters, which represents 1.66% of all parameters. I modified the LoRA configuration for the second task to increase the number of trainable parameters (doubling the r-parameter and LoRA-alpha), resulting in 20 million trainable parameters. I adjusted the batch size per device for both training and validation datasets to two to ensure better stability. During fine-tuning, each example consisted of a prompt asking the model to classify party affiliation based on the first 1,000 characters of a political speech. This simple instruction-tuning setup aligns the model more closely with the target classification task without requiring manual feature engineering.

2.3 Results

As mentioned above, we can observe that the performance of the base Qwen 3 model is inferior across all metrics compared to the fine-tuned model. This demonstrates that the fine-tuning process was effective, and through our procedure, we were able to improve accuracy by 0.27, precision by 0.31, recall by 0.225, and F1-score by 0.337. The base model essentially predicted only three different classes. Two of these classes represent the most common classes in the dataset, so it is not surprising that the model tended to classify speeches to these parties more frequently. However, it was unexpected that the model predicted Grüne as the second most frequent class after ÖVP, despite Grüne having considerably fewer speeches in the dataset. It is possible that the ÖVP and Grüne employ similar rhetoric, which may have led the model to misclassify ÖVP speeches as belonging to the Grüne party. The fine-tuned model performed better by utilizing a broader range of the party spectrum and correctly classifying significantly more speeches, including some from JETZT, which has fewer than 1,000 speeches in the dataset. The confusion matrix clearly illustrates that a major challenge for the model lies in the correct classification and distinction of the three main parties. ÖVP was frequently predicted when the actual speech was from FPÖ or SPÖ, and FPÖ and SPÖ were also confused with each other. As with the base model, this is most likely due to ÖVP having the highest number of speeches in the dataset. In retrospect, it would have been advisable to undersample the training set to account for the overrepresentation of this and other parties.

Model	**Accuracy**	**Macro Precision**	**Macro Recall**	**Macro F1 Score**
Base Qwen 3	0.290	0.383	0.200	0.143
Fine-tuned Qwen 3	0.562	0.693	0.425	0.480

Table 1: Performance comparison between base and fine-tuned Qwen 3 models (Task 1)

In the second task, the objective was to predict party orientation based on the speeches. Here, the performance of the fine-tuned model was not substantially better compared to the first task. While the base model still performed slightly worse than the fine-tuned version, I could not achieve the same level of improvement as before. With an accuracy of 0.28, the model is inadequate for this classification task. Despite modifying the LoRA configuration to increase the number of trainable parameters at the expense of efficiency, this did not enhance the model's performance. The confusion matrix reveals the underlying issue with the model. The base model predicted only one class, which is not surprising. However, unlike the fine-tuned model in Task 1, this

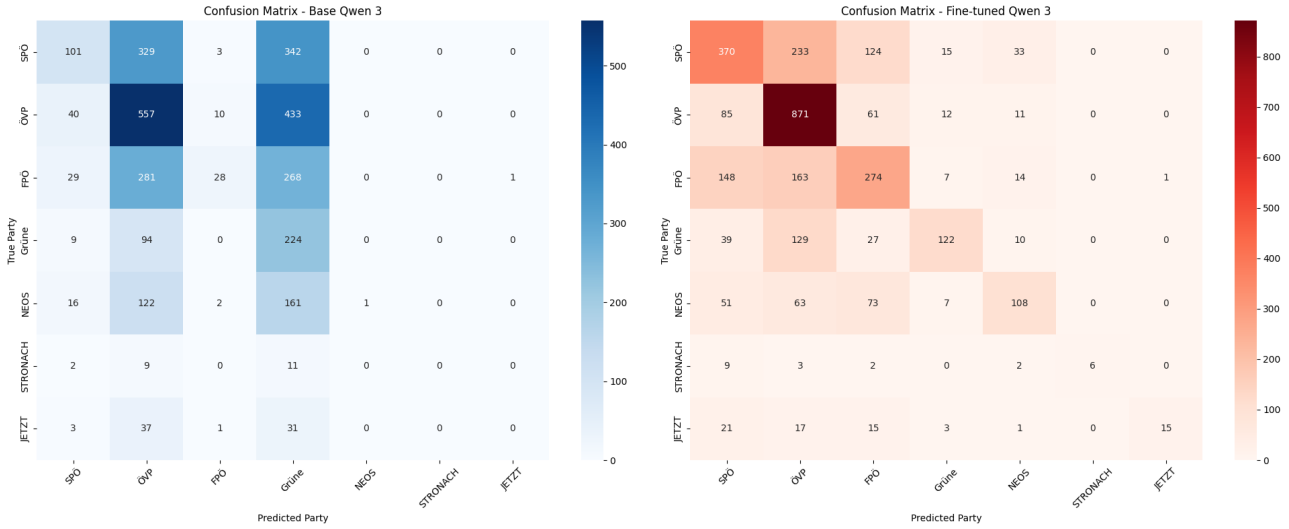


Figure 2: Confusion Matrix: Comparison of Base vs. Fine-tuned Qwen (Task 1)

fine-tuning did not lead the model to utilize more than two classes. This suggests that predicting ideology is more challenging, particularly with classes that are closely positioned on the political spectrum.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1 Score
Base Qwen 3	0.246	0.049	0.200	0.079
Fine-tuned Qwen 3_2	0.282	0.122	0.209	0.138

Table 2: Performance comparison between base and fine-tuned Qwen 3_2 models (Task 2)

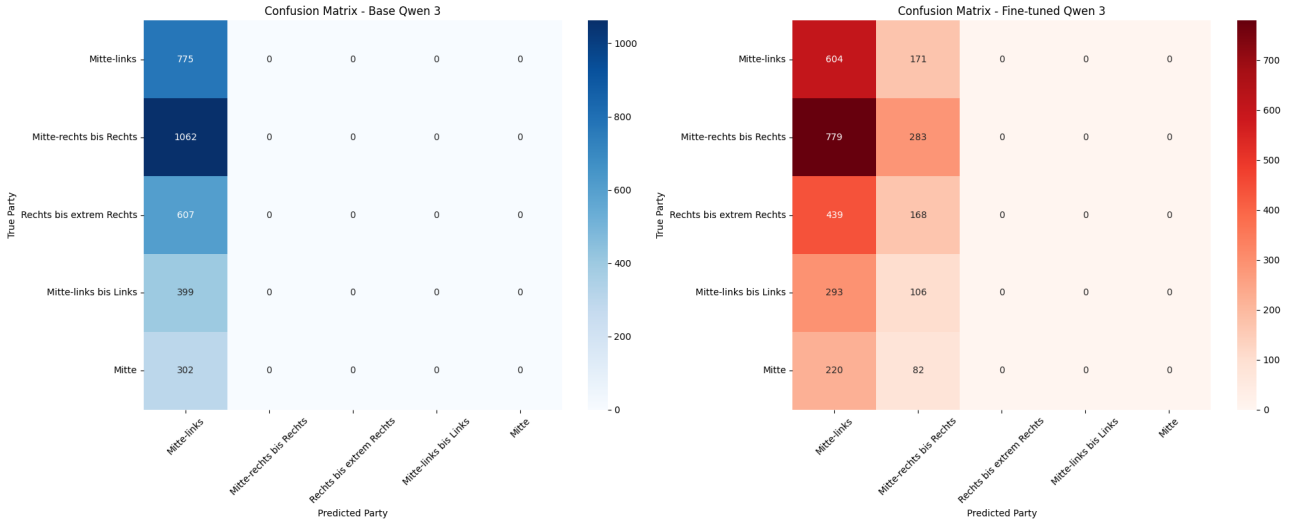


Figure 3: Confusion Matrix: Comparison of Base vs. Fine-tuned Qwen (Task 2)

3 Conclusions and Discussion

This report demonstrated that LoRA-based fine-tuning led to a substantial increase in classification performance for party prediction. The accuracy improved from 0.290 to 0.562, and the model was able to classify a wider range of parties, including smaller ones. In contrast, the second task—predicting ideological orientation—proved more challenging, and fine-tuning did not yield strong improvements. This suggests that ideological categories are more difficult to detect and may require additional context or a better-defined label structure. From

a methodological perspective, this report demonstrates that even relatively small language models such as Qwen 3 can be fine-tuned effectively using LoRA, which significantly reduces training costs by updating only approximately 1.66% of the total parameters. While tuning LoRA parameters (e.g., rank or alpha) offers some flexibility, increasing capacity alone did not guarantee better results in the second task. The methods explored in this report could be applied in several real-world contexts. First, parliamentary monitoring tools could automatically classify speeches by party or ideological leaning in near real-time, assisting journalists, researchers, or political scientists. Second, the method could support the construction of labeled datasets in low-resource settings by using the fine-tuned model for weak supervision. Despite promising results in Task 1, performance in Task 2 (ideological classification) remained low, indicating that party orientation is more difficult to infer from short speech excerpts. This could be attributed to overlapping rhetorical patterns between ideologies or insufficient label clarity. Additionally, the model tends to overpredict dominant classes, likely due to class imbalance in the training set. Addressing this issue through class weighting or undersampling could improve generalization. Finally, while LoRA is efficient, using larger base models (e.g., GPT-4) might yield better zero-shot performance.